Reg. No. | | | | | | | | | | | [18MTASD202]

SECOND SEMESTER M.TECH EXAMINATION, JUNE 2019

COMPOSITE STRUCTURES MATERIALS

Time: 03 Hours **Maximum Marks:100**

Instructions :

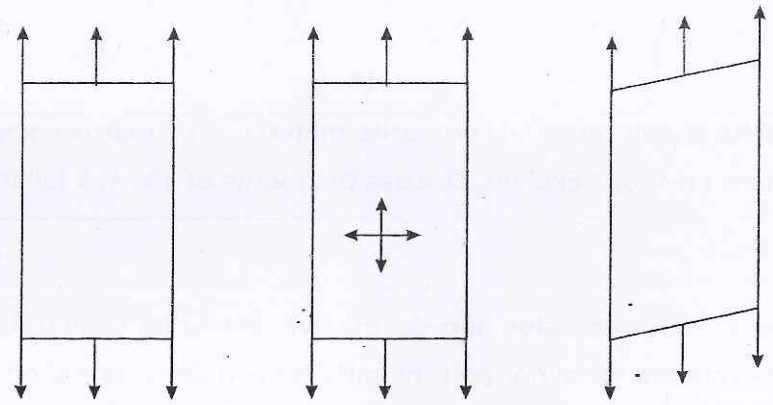
i. Answer five full questions
ii. Question 1 and 2 are compulsory
iii. Draw neat diagrams wherever necessary

1. a. Define the following **10**

i. Homogeneous and non-homogenous materials
ii. Micromechanics and macromechanics
iii. Specific strength and specific heat
iv. isotropy and orthotropy

b. What is hybrid laminate? Discuss their types **10**

2. a. Define modular ratio. Identify the type of materials based on their mechanical properties shown below. Draw their behaviour under normal and shear stresses **08**



(a) (b) (c)

b. Compute the strain components (microns) for the given stress strain $(\sigma - \varepsilon)$ relations of an orthotropic material. **12**

$$25x + 5y + z = 106.8$$

$$64x + 8y + z = 177.2$$

$$144x + 12y + z = 279.2$$

Where

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \end{bmatrix}$$

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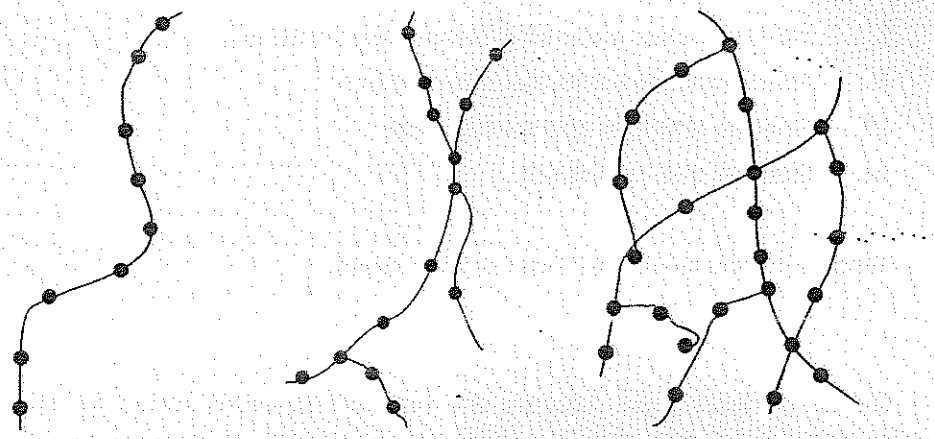
3. a. Differentiate between metals, alloys and composites **07**

b. Classify the composite materials based on the reinforcement materials with neat diagram **07**

c. Volume fraction of the fibre in the composite material is 80% and the remaining composition is made up of matrix. Estimate the elastic modulus of the fibre which has isotropic properties when maximum and minimum elastic modulus of the composite is 200GPa and 100GPa respectively. Modulus of elasticity of matrix is 5GPa. **06**

OR

4. a. Identify and explain polymers given below with suitable examples **06**



(a) (b) (c)

b. Enumerate the classification of composite material w.r.t matrix composition **07**

c. What is failure envelop? Explain. Discuss the merits of Tsa-Hill failure theory for the lamina **07**

5 a. Describe the acoustic emission non-destructive testing for composites. **10**

b. Describe the relation for minor poisons ratio for two dimensional orthotropic materials. **10**

OR

6. a. Discuss the process of manufacturing the composites by pultrusion method **10**

b. Compute the state of the strain at **10**

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

The displacement field in the lamina is given by

$$u = 10^{-5}(x^2 + 6y + 7xz)$$

$$v = 10^{-5}(yz)$$

$$w = 10^{-5}(xy + yz^2)$$

7. a. What are prepregs? Explain how the prepregs are made **10**

b. Find the strains in the 1-2 coordinate system if the Kevlar/epoxy system is loaded as shown below **10**

Take

$E_1 = 181\ GPa, \quad E_2 = 10.3\ GPa, \quad \vartheta_{12} = 0.28, \quad G_{12} = 7.17\ GPa$

OR

8. a. Discuss the radiography non-destructive testing for composites **10**

b. The three dimensional stress-strain relations for the orthotropic material is given in the Fig.8b below. Derive the two dimensional stress-strain relations and elements of reduced stiffness matrix **10**

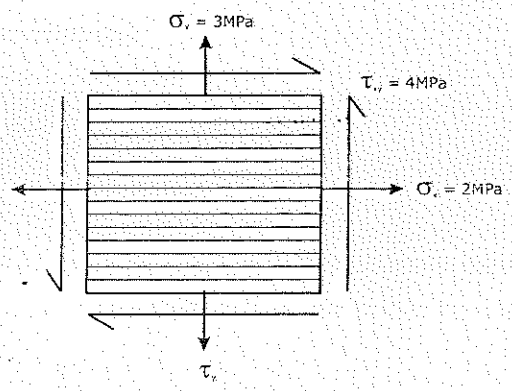


Fig. 8b